

FUTURE.STAGE by Evil Eye Pictures & Patrick Osborne

Alastair Macleod

Dan Rosen

Patrick Osborne

Yovel Schwartz

Justin Schubert

Brian W. Smith

Ben Peck

James Ritts

Andrew Angulo

alastairm@evileyepictures.com

dr@evileyepictures.com

patricko@evileyepictures.com

patricko@evileyepictures.com

justins@evileyepictures.com

brianwsmith@runbox.com

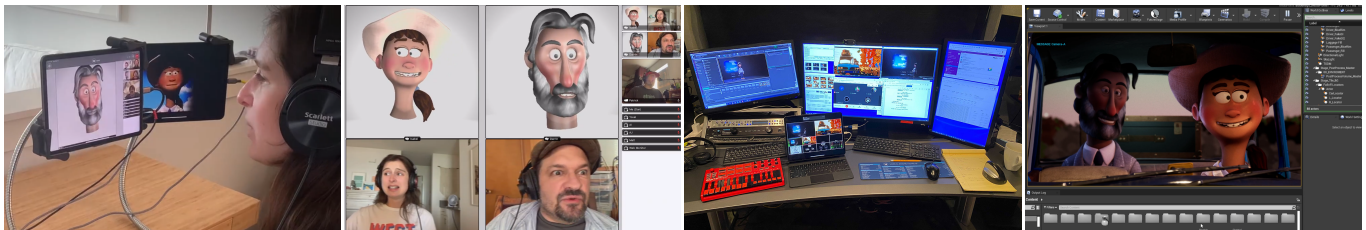
benp@evileyepictures.com

james@haibalabs.com

andrewa@evileyepictures.com

Evil Eye Pictures LLC

Berkeley, United States



ABSTRACT

In this paper, we describe a distributed collaborative system to present a real-time live computer animated show featuring more than one character and more than one interactive show element that are individually and remotely controlled each from different physical locations.

KEYWORDS

realtime, animation, motion capture, networking

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1 OVERVIEW

The motivation for this system is to facilitate the remote connection between live performers using a low entry barrier and simple interface. Integration was done with an existing real-time rendering platform and leveraged existing workflows and resources where possible. This approach maximized the potential and created new opportunities for interactive live events.

During the development of this system, it was rare for more than one person to be in the same room, despite there being as many as 10 individuals all contributing to a live performance.

2 STAGE.KIT

The primary interactive component for performers is a custom built iOS application that provides face motion capture (Apple ARKit), audio capture, video conferencing and metadata transmission features. This application, known as STAGE.KIT, locally creates a simple render of the character avatar and presents both the avatar and video

of the performer to a video conference system. An existing video conferencing platform was adopted to leverage known solutions to common communication problems such as; feedback suppression, latency and optimization. The interface was presented to the operator in a familiar and intuitive manner, similar to video conferencing platforms that are commonly used.

In addition to the video conferencing system, a separate component of STAGE.KIT timestamped and transmitted the performer's voice audio data and performance metadata to any number of remote rendering clients. This system was optimized for fidelity, synchronization and network stability with a cost to latency and data optimization.

3 STAGE.LINK

Our "STAGE.LINK" plugin enables the rendering clients to receive data from multiple remote sources and render the scene for local viewing by an operator, or for transmission to the audience using a video content distribution platform. The audio is time synchronized with the animation and spatially placed in the scene to simulate placement for a more immersive result.

4 TESTING

To test and demonstrate this system, we chose an improvised performance situation, where a performer asks the viewing audience for a suggestion and then would act out a scene where a rideshare driver picks up a passenger and is determined to keep their 5 star rating, despite the events that unfold. Environmental changes were triggered in real-time, including the addition of a hand coming out of a suitcase to give the performers cues to react to.

5 STAGE.CONTROL

The character's body animation and the suitcase were managed by an animation state machine which was remotely updated using a simple windows desktop application, known as STAGE.CONTROL. This application broadcasts control events to all render clients. More than one instances of this application can be operated at the same time.

6 EVENT TIMING

The control events were not delayed and time synchronized with the performance data, they were passed on immediately, giving the control operator a second or two advantage for triggering events. The effect of this configuration was the control operator could hear a cue from the performer in the video conference call and trigger an event which would come in time with the cue - given that the operator delay acting on the cue and the performance delay synchronization were similar. This would give the appearance that the operator is anticipating, rather than reacting to performance cues. The timing of these events took practice for the team, but was effective at giving the operator an advantage.

7 AUDIENCE INTERACTION

Finally, the audience watching the show can interact with the performers by entering text in a chat or online commenting interface that the performers can read during the show. The end-to-end

latency is only a few seconds providing opportunity for live interactions between the performers and viewing audience.

8 SUMMARY

The end result is an engaging, highly interactive, live show where all the performers can be at home, using readily available hardware. The show itself can be rendered in multiple locations and broadcast to the world.